

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

Annexure A

CONCEPT MAPPING

Code of Conduct:

I, N. Akshaya (192425129) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: N. Akshaya

98
100

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

98
100

✓

C01-AT2-Concept Map

N. Akshaya

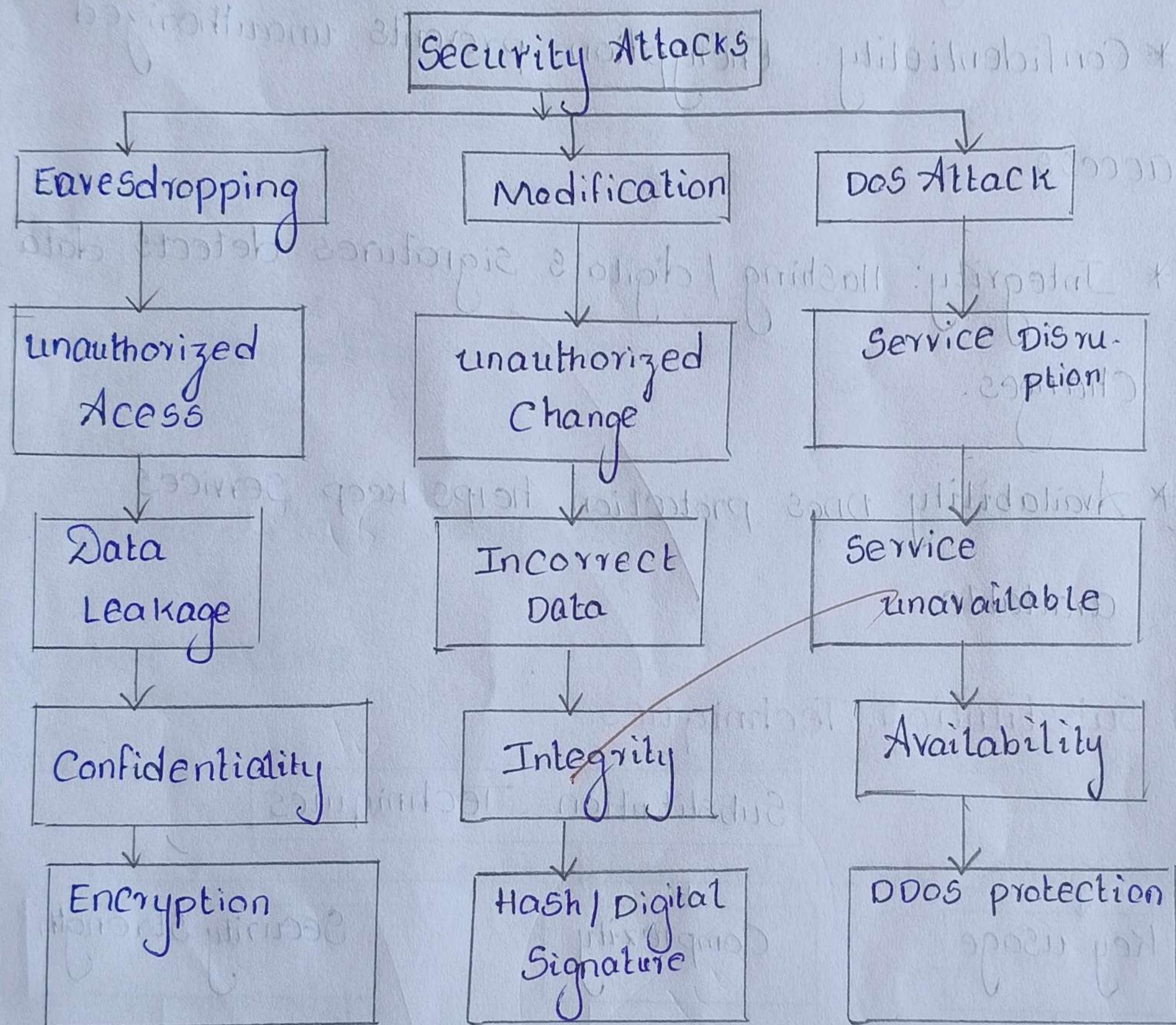
192425129

CSA5101

Cryptography and
Network Security

1) Security Attacks, Threats, Risks and Security Goals

Concept



a) Attack → Threat → Risk

* Eavesdropping → unauthorized access → Data leakage

Ex: Attacker reads private messages.

* Modification → unauthorized change → Incorrect data

Ex: Attacker change a bank transaction.

* Dos → service disruption → unavailability.

Ex: Attacker makes a website unavailable.

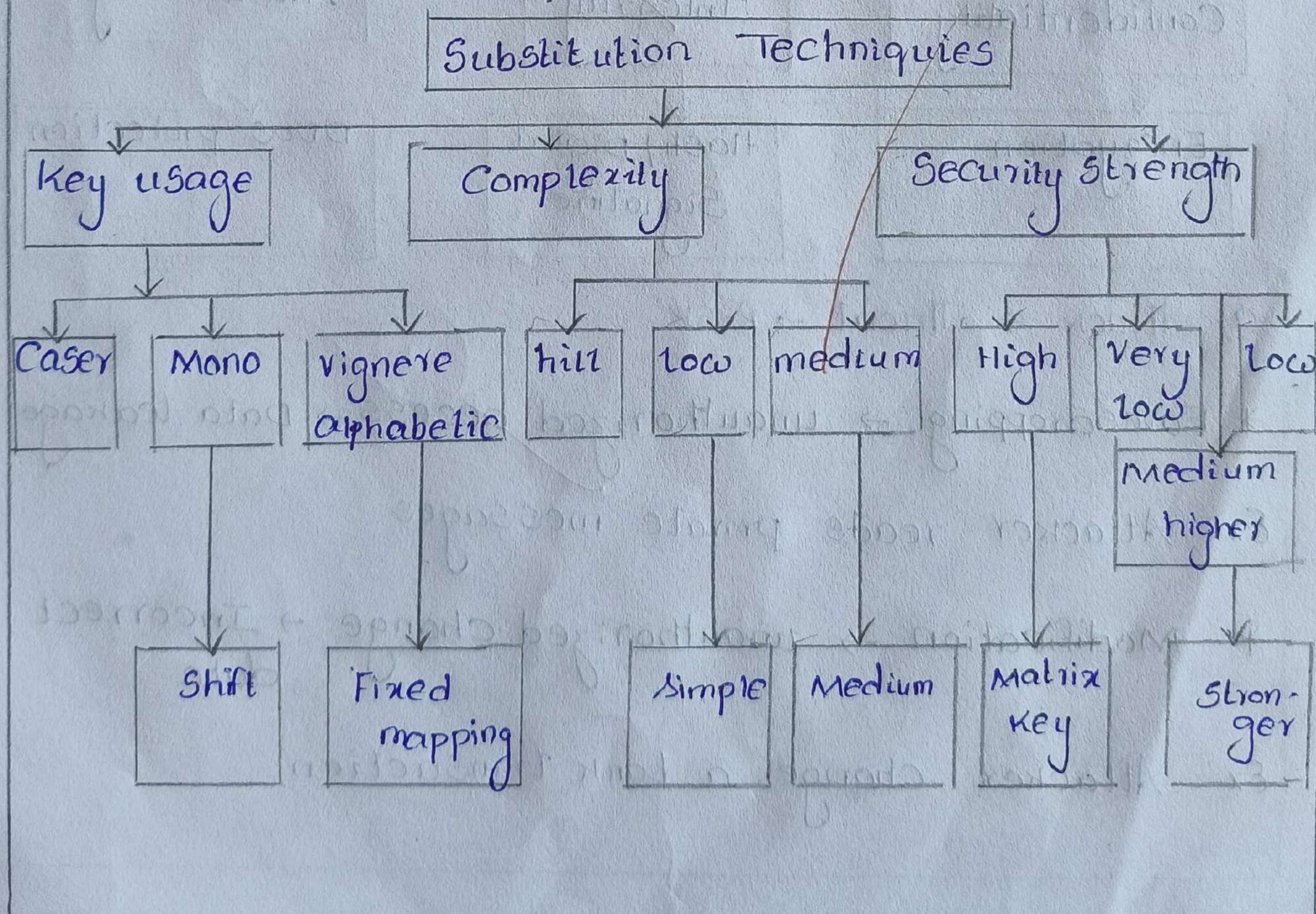
by security Goals → mitigation

* Confidentiality: Encryption prevents unauthorized access.

* Integrity: Hashing / digital signatures detects data changes.

* Availability: DDoS protection helps keep services available.

2) Substitution Techniques



a) key usage, Complexity and Security

* Caser: Fixed Shift $\rightarrow$ Low Complexity $\rightarrow$ Very low security

* Monoalphabetic:

Fixed mapping $\rightarrow$ Low Complexity $\rightarrow$ Low security

* Vignere:

Medium Complexity $\rightarrow$ Medium Security.

* Hill:

Matrix Key $\rightarrow$ high Complexity $\rightarrow$ higher security.

b) Most Secure

Hill Cipher is the strongest among these because it uses matrix based encryption and greater mathematical Complexity.

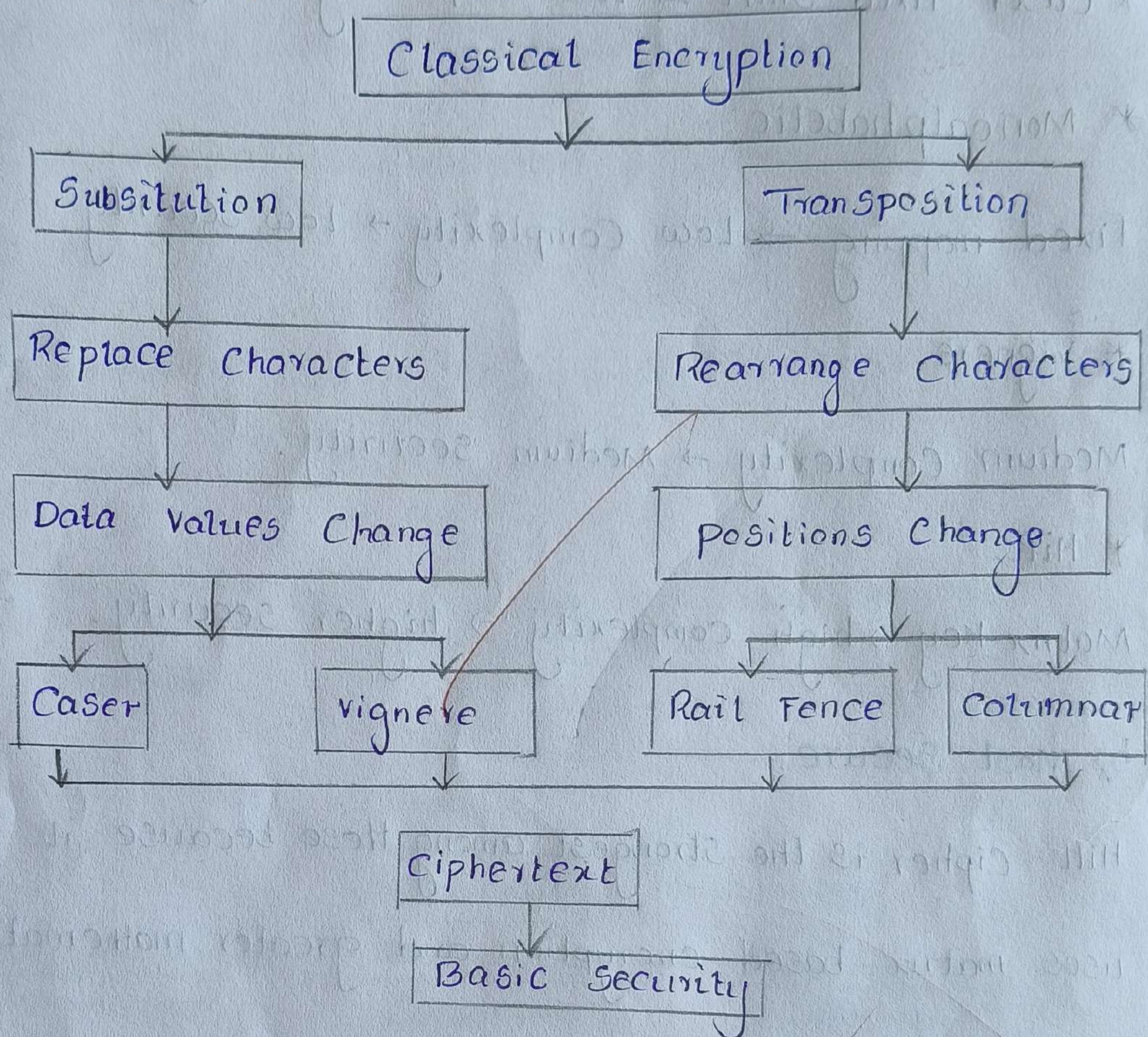
Analysis:

Security Strength generally increases from

Caser $\rightarrow$ Monoalphabetic $\rightarrow$ vignere $\rightarrow$ Hill

3) Classical Encryption Schemes

Concept Map:



a) Similarities and Differences

Similarity: Both convert plaintext into ciphertext to protect information.

Difference:

* Substitution: Replaces characters

* Transposition: Rearrange characters

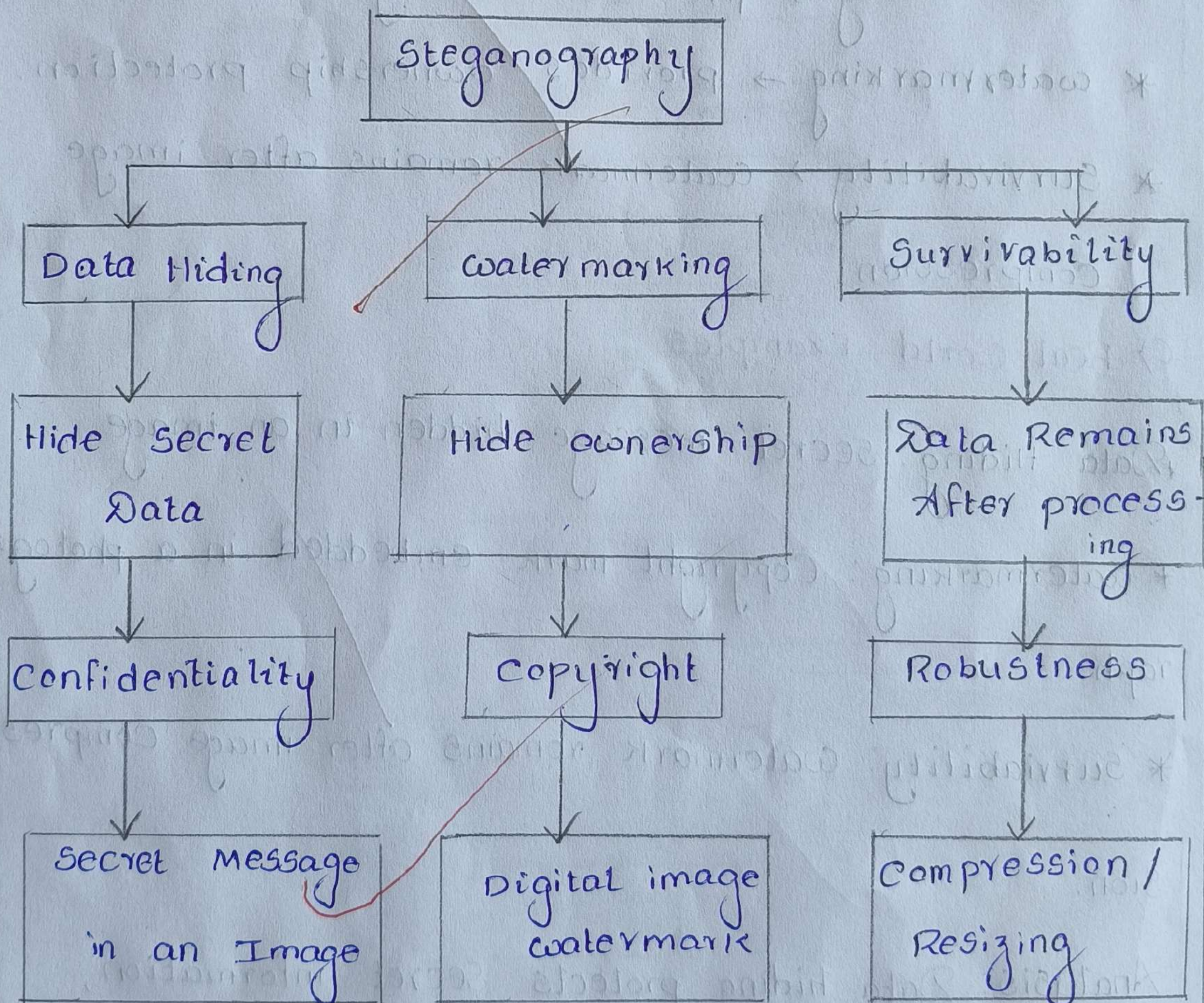
by Data structure and Security

* Substitution: Changes character values but generally keeps their positions.

* Transposition: keeps the characters but changes their positions.

4/ Steganography

Concept Map



a) Relationships

- * Data Hiding: Hides Secret information.
- * Water marking: Embeds ownership / copyright information.
- * Survivability: keeps hidden information detectable after processing.

b) Contribution to Secure Communication.

- * Data Hiding → provides secrecy.
- * Watermarking → provides ownership protection.
- * Survivability → Watermark remains after image compression.

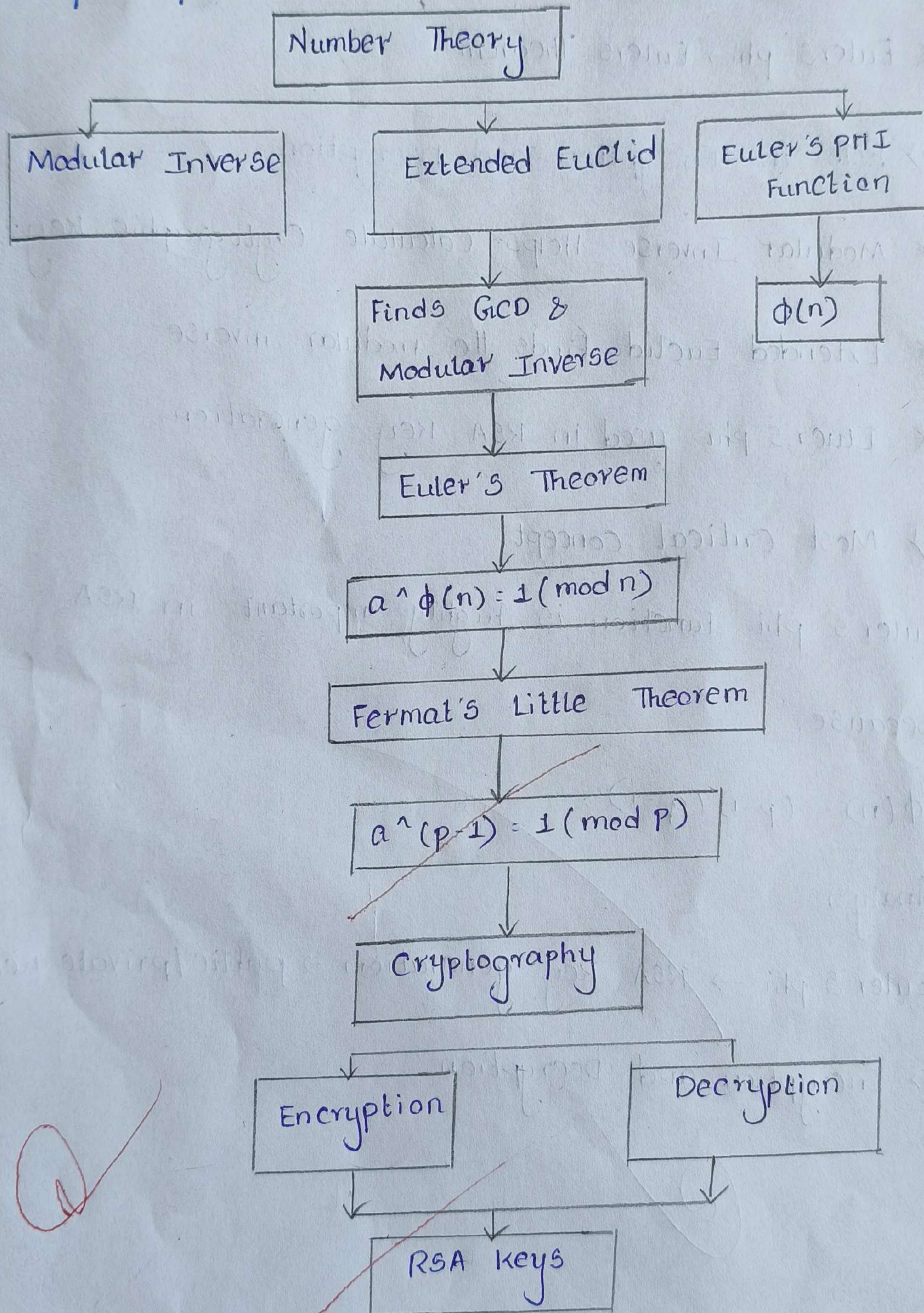
c) Real-World Examples

- * Data Hiding: secret message hidden in an image.
- * Watermarking: Copyright mark embedded in a photograph.
- * Survivability: Watermark remains after image compression.

Analysis: Data hiding protects secret information, watermarking protects ownership.

5) Number Theory Concepts in Cryptography

Concept Map



a) Mathematical connections

- * Extended Euclid $\rightarrow$ Modular Inverse
- * Euler's phi $\rightarrow$ Euler's Theorem

b) Role in Encryption and Decryption

- * Modular Inverse: Helps Calculate Cryptographic keys.
- * Extended Euclid: Finds the modular inverse.
- * Euler's phi: used in RSA key generation.

c) Most Critical Concept

Euler's phi function is highly important in RSA

because:

$$\phi(n) = (p-1)(q-1)$$

Analysis:

Euler's phi $\rightarrow$ RSA key generation $\rightarrow$ public/private keys

$\rightarrow$ Encryption and Decryption.

